

The given functions $f_1(x) = 5$, $f_2(x) = \cos^2 x$, $f_3(x) = \sin^2 x$ are linearly independent.

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Question No: 5

Marks: 2

A set of functions whose wronskian is zero then set of functions may or may not be dependent.

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Question No: 6

Marks: 10

(a) Define separable form. Just separate the variables of the given differential equation.

$$\sec y \frac{dy}{dx} + \sin(x - y) = \sin(x + y)$$

Solution

The differential equation of the form $\frac{dy}{dx} = f(x, y)$ is called **separable** if it can be written in the form $\frac{dy}{dx} = h(x)g(y)$

$$\sec y \frac{dy}{dx} + \sin(x - y) = \sin(x + y)$$

$$\sec y \frac{dy}{dx} = \sin(x + y) - \sin(x - y)$$

$$\sec y \frac{dy}{dx} = \sin(x + y) - \sin(x - y)$$

$$\sec y \frac{dy}{dx} = \sin x \cos y + \cos x \sin y - \sin x \cos y + \cos x \sin y$$

$$\sec y \frac{dy}{dx} = 2 \cos x \sin y$$

$$\frac{dy}{2 \cos y \sin y} = \cos x dx$$

$$\frac{dy}{\sin 2y} = \cos x dx$$

$$\operatorname{cosec} 2y dy = \cos x dx$$

(b) Check whether the given differential equation is exact or not if not then make it exact also show that it is exact **(Just make the equation exact do not solve it further)**.

$$e^x dx + (e^x \cot y + 2y \operatorname{cosec} y) dy = 0$$

Solution